

**SOLNETHNAYA, Russian Federation**

WMO: 305370

Lat: 54.030N

Lon: 108.255E

Elev: 1521

StdP: 13.91

Time Zone: 8.00 (E08)

Period: 90-05

WBAN: 99999

**Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -18.5      | -13.7 | -26.9                         | 1.3 | -13.8 | -22.7 | 1.7 | -9.3 | 37.6                  | 8.0  | 31.4 | 7.2  | 4.2                   | 300  | 0.739 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.2                   | 69.2            | 60.1 | 66.7 | 58.8 | 64.6 | 57.4 | 62.2                | 66.4 | 60.5 | 64.8 | 59.0 | 63.1 | 5.9                  | 290  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 60.5                            | 83.2 | 64.9 | 58.6 | 77.6 | 63.0 | 57.0 | 73.3 | 61.6 | 28.8          | 66.4 | 27.6 | 65.1 | 26.5 | 63.2 | 70.5              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 34.3              | 27.6 | 22.5 | -21.1                      | 77.5 | 5.1                | 3.5 | -24.8                                              | 80.0 | -27.7      | 82.1 | -30.6      | 84.1 | -34.3      | 86.6 |
|                   |      |      | -21.0                      | 65.1 | 5.2                | 2.4 | -24.8                                              | 66.9 | -27.9      | 68.3 | -30.8      | 69.7 | -34.6      | 71.4 |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 31.0   | 2.1   | 3.1   | 14.1  | 29.7  | 40.5  | 48.7  | 55.2  | 57.8  | 47.0  | 35.7  | 22.0  | 14.7  |
|                                                                        | DBStd       | 20.24  | 10.27 | 9.84  | 10.29 | 7.44  | 5.61  | 4.67  | 5.71  | 4.56  | 5.33  | 6.98  | 9.62  | 7.99  |
|                                                                        | HDD50       | 7412   | 1486  | 1313  | 1113  | 610   | 296   | 79    | 14    | 5     | 119   | 444   | 839   | 1094  |
|                                                                        | HDD65       | 12399  | 1951  | 1733  | 1578  | 1060  | 758   | 489   | 307   | 228   | 539   | 908   | 1289  | 1559  |
|                                                                        | CDD50       | 495    | 0     | 0     | 0     | 0     | 2     | 40    | 176   | 246   | 30    | 1     | 0     | 0     |
|                                                                        | CDD65       | 7      | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 3     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 21     | 0     | 0     | 0     | 0     | 0     | 0     | 9     | 12    | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 7.1    | 6.4   | 5.1   | 6.0   | 7.6   | 6.5   | 4.8   | 4.1   | 5.5   | 8.1   | 8.9   | 11.7  | 9.9   |
| (15) Precipitation                                                     | PrecAvg     | 15.8   | 0.7   | 0.3   | 0.4   | 0.7   | 1.2   | 1.8   | 2.8   | 2.6   | 1.7   | 0.9   | 1.3   | 1.3   |
|                                                                        | PrecMax     | 21.1   | 2.4   | 0.6   | 1.2   | 1.5   | 2.2   | 3.9   | 6.1   | 4.4   | 3.3   | 1.9   | 2.1   | 2.3   |
|                                                                        | PrecMin     | 10.4   | 0.2   | 0.1   | 0.1   | 0.3   | 0.4   | 0.6   | 0.8   | 1.0   | 0.6   | 0.2   | 0.6   | 0.7   |
|                                                                        | PrecStd     | 2.3    | 0.4   | 0.1   | 0.2   | 0.3   | 0.5   | 0.8   | 1.3   | 0.9   | 0.7   | 0.4   | 0.4   | 0.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 31.1  | 32.9  | 43.0  | 52.9  | 60.1  | 67.2  | 74.4  | 74.6  | 63.3  | 56.7  | 44.9  | 35.7  |
|                                                                        |             | MCWB   | 25.7  | 26.4  | 35.2  | 42.6  | 48.2  | 54.3  | 61.5  | 63.2  | 56.2  | 46.4  | 37.5  | 30.2  |
|                                                                        | 2%          | DB     | 26.4  | 26.9  | 37.2  | 47.1  | 55.8  | 62.9  | 70.4  | 68.9  | 60.6  | 50.8  | 41.0  | 30.8  |
|                                                                        |             | MCWB   | 22.6  | 22.4  | 30.5  | 37.8  | 45.2  | 52.5  | 60.8  | 60.3  | 53.4  | 43.7  | 34.5  | 26.0  |
|                                                                        | 5%          | DB     | 21.9  | 22.1  | 33.1  | 43.3  | 52.6  | 59.8  | 67.4  | 66.8  | 58.4  | 48.2  | 38.4  | 27.6  |
|                                                                        |             | MCWB   | 18.7  | 18.8  | 27.7  | 35.5  | 42.7  | 50.6  | 59.3  | 59.2  | 52.1  | 41.8  | 32.6  | 23.4  |
|                                                                        | 10%         | DB     | 16.4  | 17.6  | 29.2  | 40.2  | 49.8  | 57.2  | 65.0  | 65.0  | 56.2  | 45.7  | 34.9  | 25.2  |
|                                                                        |             | MCWB   | 13.6  | 14.9  | 24.8  | 33.4  | 41.0  | 49.0  | 58.0  | 58.0  | 50.1  | 39.5  | 29.7  | 21.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 26.1  | 26.3  | 35.9  | 43.1  | 49.4  | 56.6  | 65.9  | 64.8  | 59.1  | 48.1  | 38.2  | 30.7  |
|                                                                        |             | MCDB   | 30.0  | 31.9  | 42.3  | 52.6  | 58.9  | 62.6  | 71.5  | 71.5  | 63.0  | 54.0  | 43.6  | 34.7  |
|                                                                        | 2%          | WB     | 22.6  | 22.4  | 31.2  | 38.6  | 46.3  | 54.0  | 62.8  | 62.3  | 55.2  | 44.8  | 34.9  | 26.4  |
|                                                                        |             | MCDB   | 25.8  | 26.3  | 36.8  | 46.0  | 54.4  | 60.8  | 67.2  | 66.4  | 59.2  | 50.2  | 40.1  | 30.0  |
|                                                                        | 5%          | WB     | 18.6  | 18.8  | 27.8  | 35.9  | 43.8  | 52.0  | 60.7  | 61.0  | 52.9  | 42.6  | 32.7  | 23.8  |
|                                                                        |             | MCDB   | 21.8  | 21.9  | 32.7  | 42.2  | 50.9  | 58.5  | 65.5  | 65.1  | 57.2  | 47.6  | 38.0  | 27.2  |
|                                                                        | 10%         | WB     | 13.8  | 15.0  | 24.9  | 33.8  | 41.7  | 50.0  | 58.8  | 59.6  | 50.8  | 40.0  | 30.2  | 21.8  |
|                                                                        |             | MCDB   | 16.2  | 17.5  | 29.2  | 39.9  | 48.7  | 56.0  | 64.1  | 63.4  | 55.2  | 44.9  | 34.9  | 25.2  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.4  | 13.0  | 14.0  | 13.2  | 14.0  | 14.2  | 13.5  | 11.2  | 10.7  | 11.3  | 10.2  | 9.5   |
|                                                                        |             | MCDBR  | 16.2  | 17.7  | 17.9  | 18.2  | 19.1  | 18.4  | 18.1  | 13.0  | 13.1  | 14.2  | 13.5  | 11.2  |
|                                                                        |             | MCWBR  | 14.0  | 14.3  | 13.2  | 12.6  | 11.9  | 11.3  | 10.6  | 8.7   | 10.2  | 10.6  | 11.0  | 9.7   |
|                                                                        | 5% WB       | MCDBR  | 16.3  | 17.3  | 16.9  | 17.2  | 18.2  | 17.3  | 15.1  | 11.3  | 11.9  | 13.8  | 13.2  | 10.9  |
|                                                                        |             | MCWBR  | 14.1  | 14.1  | 12.8  | 12.4  | 12.4  | 11.8  | 10.4  | 8.9   | 10.5  | 11.1  | 11.1  | 9.8   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.272 | 0.305 | 0.332 | 0.380 | 0.430 | 0.464 | 0.447 | 0.404 | 0.351 | 0.317 | 0.285 | 0.268 |
|                                                                        | taud        |        | 2.024 | 1.993 | 2.025 | 2.113 | 2.110 | 2.117 | 2.221 | 2.323 | 2.416 | 2.326 | 2.070 | 1.984 |
|                                                                        | Ebn at Noon |        | 206   | 239   | 263   | 266   | 259   | 251   | 253   | 258   | 259   | 241   | 199   | 176   |
|                                                                        | Edh at Noon |        | 23    | 34    | 41    | 43    | 46    | 47    | 42    | 35    | 28    | 24    | 22    | 20    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 311   | 628   | 1079  | 1457  | 1737  | 1912  | 1737  | 1455  | 1020  | 583   | 292   | 206   |
|                                                                        | RadStd      |        | 26    | 37    | 53    | 105   | 119   | 120   | 150   | 67    | 80    | 61    | 28    | 20    |

**Historical Trends**

|                             | DBAvg | Heating |       |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |       | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)   | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (9 neighbors) | N/A   | -2.79   | -3.62 | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +79   | N/A   |

Nomenclature: See separate page